

1. Administrative & Study Identification

Full Study Title: LIGHTBEAM-U01 Substudy 01A: A Phase 1/2 Substudy to Evaluate the Safety and Efficacy of Zilovertamab Vedotin in Pediatric and Young Adult Participants With Hematologic Malignancies or Solid Tumors **Short Title/Acronym:** LIGHTBEAM-U01A (MK-9999-01A) **Protocol Number:** MK-9999-01A **NCT Number:** NCT06395103 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck & Co., Inc.) **Investigational Product:** Zilovertamab Vedotin (ZV) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of zilovertamab vedotin, determine the preliminary recommended phase 2 dose (RP2D), and assess the objective response rate (ORR). **Secondary Objectives:** To evaluate pharmacokinetics (PK), immunogenicity, duration of response (DOR), and eligibility for transplant/CAR-T therapy for patients with B-ALL or DLBCL/Burkitt lymphoma.

2.2 Background & Rationale To address the gap in treatment options for pediatric and young adult patients with relapsed or refractory hematologic malignancies and solid tumors. ROR1 is an oncofetal protein expressed in various blood and solid cancers. Zilovertamab vedotin (ZV) is an antibody-drug conjugate (ADC) comprising a monoclonal antibody against ROR1, a proteolytically cleavable linker, and monomethyl auristatin E (MMAE). This study aims to provide targeted therapy for patients who have exhausted therapies of known clinical benefit.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basket Study) **Primary Purpose:** Treatment **Intervention Model:** Single Group Assignment **Control Type:** Single-arm **Blinding:** Open-label (None) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 50 to 90 participants **Number of Sites:** Global, including ~66 locations across the United States, United Kingdom, and over 20 other countries **Estimated Study Start:** 16-Aug-2024 **Estimated Primary Completion:** 31-Mar-2029 **Estimated Study Completion:** 31-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 0 months to ≤ 25 years (Pediatric participants aged 0 to <18 years; young adults aged 18-25 years depending on the cohort). Sex: All. 2. Confirmed diagnosis of relapsed/refractory B-cell acute lymphoblastic leukemia (B-ALL), diffuse large B-cell lymphoma (DLBCL)/Burkitt lymphoma, neuroblastoma, or Ewing sarcoma. 3. For B-ALL or DLBCL/Burkitt lymphoma: Relapsed after ≥ 2 prior lines of therapy; B-ALL patients must have ≥ 5% bone marrow blasts. For neuroblastoma/Ewing sarcoma: Refractory to frontline therapy with measurable disease.

4.2 Exclusion Criteria 1. History of solid organ transplant, ongoing Grade >1 peripheral neuropathy, demyelinating form of Charcot-Marie-Tooth disease, or diagnosed with Down syndrome. 2. Ongoing graft-versus-host disease (GVHD) of any grade or receiving systemic GVHD treatment/prophylaxis. 3. Received prior radiotherapy within 4 weeks of the start of study intervention, or ongoing chronic corticosteroid therapy exceeding 10 mg daily of prednisone equivalent.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: ZV administered at a starting dose of 2 mg/kg via Intravenous (IV) infusion every 3 weeks (Q3W), escalating to 2.25 and 2.5 mg/kg or de-escalating to 1.75 mg/kg in Part 1 per a modified toxicity probability interval design-2. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Cycles repeat every 3 weeks until disease progression, unacceptable toxicity, or other discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Prophylactic intrathecal chemotherapy and/or cytoreductive therapy with steroids/hydroxyurea prior to first dose. Killed vaccines are allowed. **Prohibited:** Strong CYP3A4 inhibitors (within 7 days) or strong CYP3A4 inducers (within 14 days) before the start of study intervention; live or live-attenuated vaccines within 30 days before the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Eligibility Criteria Check, Disease Assessment (Imaging/Bone Marrow), Safety Labs
Baseline	Cycle 1, Day 1	First Dose of Zilovetamab Vedotin, PK/PD Blood Sampling, Baseline Vitals	
Interim	Every 3 Weeks	Q3W IV Dosing. Disease Assessments: Q8W for 6 months, then Q12W to 24 months (Solid tumors/DLBCL); End of each cycle for B-ALL.	
End of Study	+30 to +90 Days		

AE monitoring \$\le\$ 30\$ days after last dose (90 days for SAEs). Long-term follow-up annually. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Determination of the Recommended Phase 2 Dose (RP2D) (Part 1); Incidence of Adverse Events graded per NCI CTCAE v5.0; Objective Response Rate (ORR). **Measurement Method:** Clinical safety assessments and radiographic/bone marrow response criteria (e.g., RECIST v1.1 for solid tumors, WHO/IPNHL criteria for lymphomas).

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetics (PK) and Immunogenicity profiles. 2. Duration of Response (DOR) and determination of eligibility for subsequent transplant or CAR-T therapy for patients with B-ALL or DLBCL/Burkitt lymphoma.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs will be monitored through 30 days after the last dose of study treatment (or 30 days if a new anticancer therapy is initiated) and graded per NCI CTCAE v5.0. **Serious Adverse Events (SAEs):** Monitored up to 90 days after the last dose of study treatment. **Data Monitoring Committee (DMC):** An internal and/or independent data monitoring committee will oversee the dose escalation and confirmation (Part 1) prior to efficacy expansion (Part 2).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (ORR); Safety Set for all patients receiving at least one dose of ZV. **Sample Size Justification:** Approximately 50-90 participants across 4 disease cohorts. Part 1 will enroll 3-12 patients per dose level. In Part 2, 10 patients will be enrolled in each cohort. **Handling of Missing Data:** Missing data will be handled per sponsor standard statistical methodologies for early-phase oncology trials, generally evaluating responses based on available radiographic and clinical evaluations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of clinical data, dose-limiting toxicities (DLTs), and adherence to the modified toxicity probability interval design-2 algorithm. **Audit Trail:** Comprehensive eCRF documentation, query resolution

workflows, and central review of radiographic and pathology findings.

11. Study Identifiers & Governance

Primary ID: MK-9999-01A **Secondary IDs:** NCI-2024-04094, 2023-507178-41-00, jRCT2031250824, LIGHTBEAM-U01 **Registry Number:** NCT06395103 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** LIGHTBEAM-U01A (Substudy 01A) **Official Regulatory Title:** LIGHTBEAM-U01 Substudy 01A: A Phase 1/2 Substudy to Evaluate the Safety and Efficacy of Zilovertamab Vedotin in Pediatric and Young Adult Participants With Hematologic Malignancies or Solid Tumors

12. Clinical Framework (Oncology Specific)

Note: This section from the template is N/A for this oncology trial but is retained for template structural integrity. Replaced with Oncology Specific Framework. **Primary Condition:** Relapsed/Refractory B-ALL, DLBCL/Burkitt Lymphoma, Neuroblastoma, Ewing Sarcoma **Diagnosis Threshold:** Relapsed after ≥ 2 prior lines of therapy (hematologic) or refractory to frontline therapy (solid tumors) **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) **Optimization Target:** Objective response and bridging to cellular therapies (Transplant/CAR-T)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical trial pathway for relapsed/refractory disease. **Education Protocol (HCP):** Investigator meeting and Site Initiation Visits (SIV) covering the protocol, dose-escalation rules, and ZV specific toxicity management. **Education Protocol (Patient):** Informed consent process, counseling on potential ADC-related adverse events. **Quality Audit Mechanism:** Routine sponsor audits and continuous safety reviews during dose-escalation cohorts.

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years (and beyond for survival follow-up). **Onsite Visit Cadence:** Every 3 weeks (Q3W) for treatment administration. **Remote Monitoring Frequency:** Continuous safety monitoring reporting. **Checklist Review Interval:** Disease assessments Q8W for the first 6 months, then Q12W through 24 months, then Q24W through 5 years.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					